

MOHAMED EID SHEHATA

Data Scientist | AI Engineer

Egypt | mohamedshehata210006@gmail.com | +20 111 645 5176 | [LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with hands-on experience building and deploying Machine Learning and Deep Learning systems across predictive modeling, NLP, computer vision, and Retrieval-Augmented Generation (RAG) pipelines. Track record includes models reaching up to 98.6% accuracy and informing decisions tied to \$15.8M+ in business revenue. Proficient in Python, SQL, TensorFlow, PyTorch, and Scikit-learn, with strong expertise in statistical analysis, feature engineering, model evaluation, and Explainable AI (XAI). Experienced in building BI dashboards (Power BI, Tableau) and deploying production-grade ML applications with Streamlit.

CORE TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** Classification, Regression, Clustering, CNNs, RNNs, BiLSTM, Transfer Learning, Hyperparameter Tuning, Model Evaluation & Validation
- **Computer Vision:** Image Classification, Object Detection & Segmentation (YOLO, Detectron2), OpenCV, Transfer Learning, CNNs
- **NLP & LLMs:** Text Preprocessing, Sentiment Analysis, NLP Pipelines, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering
- **Programming & Frameworks:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras), SQL (MySQL, SQL Server)
- **Data Analytics & BI:** Power BI (DAX, Data Modeling, ETL), Tableau, KPI Design, Data Visualization & Storytelling, Exploratory Data Analysis (EDA)
- **Explainable AI & Interpretability:** SHAP, Feature Importance Analysis, Profit-Based Scoring Models
- **Data Engineering & Deployment:** ETL Pipeline Development, Data Cleaning & Transformation, Streamlit App Deployment, Model Productionization, API Integration
- **Cloud platforms:** Azure
- **Core Competencies:** Predictive Modeling, Statistical Analysis, A/B Testing Principles, Business Intelligence, Data Storytelling, Cross-Functional Communication

KEY PROJECTS

Deep Learning Surrogate Models for Solid Rocket Motor Grain Design (Graduation Project) 2026 | Tools: Python, TensorFlow, Keras, CNN, LSTM/GRU, Monte Carlo Dropout, Streamlit [Link](#)

- Led the design and engineering of 18 deep learning pipelines (9 forward + 9 inverse), replacing CFD simulation with millisecond-level neural inference for rocket motor grain design.
- Achieved up to 98.6% R^2 on curve prediction and sub-2% MAPE on geometric recovery by developing custom CNN, GRU/LSTM, and cross-attention architectures with Monte Carlo Dropout uncertainty quantification.

Road Damage Detection (Object Detection & Model Benchmarking) 2026 | Tools: Python, YOLO (Ultralytics), ONNX, Streamlit, OpenCV [Link](#)

- Built a real-time road-defect detection system covering 3 classes (Pothole, Drain Hole, Sewer Cover) from a custom-labeled video dataset.
- Benchmarked 3 YOLO architectures (YOLOv8n, YOLO11n, YOLO26n) on mAP50, mAP50-95, Precision/Recall, FPS, and model size, selecting YOLOv8n (89.8% mAP50, 89.4% Precision, ~104 FPS) as the optimal accuracy-speed tradeoff.
- Trained the final model for 100 epochs with early stopping, exported it to ONNX for faster inference, and deployed it in an interactive Streamlit app that processes uploaded videos and auto-generates damage reports.

Oral Disease Image Classification (CNN & Transfer Learning) 2026 | Tools: Python, TensorFlow, ResNet50, MobileNetV2, EfficientNetB0, Gradio [Link](#)

- Built a 6-class dental condition classifier (Calculus, Caries, Gingivitis, Mouth Ulcer, Tooth Discoloration, Hypodontia) from a raw, inconsistent dataset by engineering a cleaned, leak-free, stratified 80/10/10 train/val/test split.
- Benchmarked 4 architectures head-to-head (custom CNN, MobileNetV2, ResNet50, EfficientNetB0), applying model-specific preprocessing pipelines to ensure a fair comparison across transfer-learning models.

- Selected ResNet50 as the top performer (~87% validation accuracy, smallest train/val gap), and deployed it as a live, interactive Gradio web app on Hugging Face Spaces for real-time predictions with confidence scores.

Amazon Reviews Sentiment Analysis (NLP / Deep Learning) 2025 | Tools: Python, TensorFlow, Keras, RNN, BiLSTM, NLP [Link](#)

- Achieved 94.58% accuracy and 0.9860 ROC-AUC by engineering a BiLSTM-based deep learning model through systematic hyperparameter tuning and early stopping.
- Delivered a production-grade sentiment classification system, enabling reliable, scalable business intelligence for e-commerce review analysis.

Customer Churn & Profit Optimization (ML + Explainable AI + BI) 2024 | Tools: Python, Scikit-learn, SHAP, Streamlit, Power BI [Link](#)

- Developed an end-to-end, revenue-driven machine learning pipeline addressing a 26.5% customer churn rate and \$3.68M in revenue exposure.
- Reduced false negatives by optimizing model recall and AUC, ensuring proactive identification of high-risk customers before churn occurred.
- Engineered a custom Profit-Based Priority Score (Probability × CLTV) to maximize retention ROI and guide budget allocation.
- Deployed an Explainable AI (XAI) Streamlit application with real-time SHAP explanations alongside a companion Power BI dashboard, surfacing ~\$4.4K in tracked insights.

TECHNICAL STRENGTHS

- Building end-to-end AI pipelines spanning data ingestion, modeling, and deployment-ready solutions.
- Designing interactive BI dashboards and KPI frameworks that directly inform business decisions.
- Applying NLP and Generative AI techniques for text analytics, sentiment analysis, RAG systems, and intelligent applications.
- Designing, training, and tuning deep learning and computer vision architectures (CNNs, RNNs/BiLSTM, object detection).
- Translating complex, ambiguous business problems into data-driven AI products and analytics solutions.

CERTIFICATIONS

- AI and Data Science Diploma — Instant Software Solutions
- Associate Data Scientist — DataCamp
- Machine Learning — Coursera
- AI Foundations Associate — Oracle
- Data Science & Machine Learning — Edraak
- Google Data Analytics — Coursera
- Data Analyst Track (Python, Power BI, Tableau) — DataCamp
- SQL & Advanced SQL — 365 Data Science

EDUCATION

Bachelor's Degree in Navigation Science & Space Technology

Beni Suef University, Egypt — 2021 – 2026

LANGUAGES

Arabic (Native) | English (Intermediate)